

Weighted Mask Survival and Sharp Projective-to-Atomic Distortion for Residual Determinant Packets

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July 2026

Abstract

The preceding twin-prime-correlation (TPC) reduction controls the complete frozen actual-coefficient residual family by a signed projective row-square envelope, whereas the inherited normalization is the physical atomic diagonal. We identify the exact finite-dimensional distortion between these two geometries. For a Hilbertized coefficient tensor T , the optimal rank-one projective envelope is its nuclear norm and the atomic energy is its squared Frobenius norm. The intrinsic energy loss is therefore the nuclear participation rank

$$r_{\text{nuc}}(T) = \frac{\|T\|_{S_1}^2}{\|T\|_{S_2}^2}.$$

For a specified signed representation there is a second, independent factor: its cancellation condition number. We prove the exact product formula for the two losses and show that cancelling layer pairs make every comparison uniform over all exact representations impossible.

For a weighted row mask $A = D_a K D_b$, the bounded projective norm used in TPC-36 transfers sharply to the Hilbert geometry:

$$\|A\|_{S_1} \leq \|K\|_{\pi, \infty} \|a\|_2 \|b\|_2, \quad r_{\text{nuc}}(A) \leq \min \left\{ \text{rank } A, \frac{\|K\|_{\pi, \infty}^2}{\delta_w} \right\},$$

where $\delta_w = \|A\|_{S_2}^2 / (\|a\|_2^2 \|b\|_2^2)$ is the weighted survival parameter (the weighted admitted fraction for a 0-1 mask). Permutation masks attain the density bound exactly, so bounded projective cost one does not by itself imply any dimension-free atomic estimate. Ferrers masks, by contrast, have distortion $(2/\pi^2)(\log n)^2 + O(\log n)$.

We then study the complete-universe extension of the literal static TPC mask formula. Standard row counts show that every row and column has admitted degree $(1 - o(1))N$. Combined with a specified explicit TPC-36 projective representation, this gives only $X^{o(1)}$ energy loss for that complete formula model. Nevertheless the same admitted support contains a perfect-matching permutation packet with distortion $N = X^{267/400 + o(1)}$; it is a support-class witness, not the literal physical coefficient. Thus static mask geometry is soft on the full row cell, while support information alone is provably insufficient.

If the entire TPC endpoint allowance is assigned to the isolated Hilbertized face, its exact fixed-representation target is $c_{\mathcal{R}}^2 r_{\text{nuc}} \leq X^{1/400 + o(1)}$. This is not an if-and-only-if criterion for full TPC-48 closure, whose other gates may consume part or all of the same budget. The paper does not identify the full Banach profile envelope with a nuclear norm and does not compare the completed coherent prime square with the fixed- h_0 physical diagonal. Weighted active-cell density, representation-frame efficiency, coherent prime cancellation, residual grouping, and shift localization remain separate gates. No parity-barrier breach or twin-prime conclusion is claimed.

Keywords: projective tensor norm; nuclear norm; Frobenius norm; effective rank; determinant equation; row masks; Möbius correlation; parity problem.

Convention. All sets and matrices in the abstract theory are finite. The Schatten norms are denoted by $\|\cdot\|_{S_1}$ and $\|\cdot\|_{S_2}$; the latter is the Frobenius norm. Rank-one tensors $x \otimes \bar{y}$ are identified with operators $z \mapsto \langle z, y \rangle x$. Constants may depend on the fixed shift and dyadic support constants.

MSC 2020: Primary 46B28, 47B10; Secondary 11N35, 15A18, 42A05.

1 Introduction

The all-band theorem of TPC–48 leaves a precise normalization problem. The frozen residual determinant family built from the literal h_0 -coefficient field satisfies, schematically,

$$\|\widehat{\mathcal{F}}^{\text{fr}}(\alpha)\| \leq L(\alpha)^{1/2} \mathfrak{S}_{\text{pr}}(\alpha), \quad (1)$$

where L is an explicit common-profile large-sieve factor and $\mathfrak{S}_{\text{pr}}$ is the pointwise infimum of signed projective row-square envelopes [8]. The inherited TPC denominator, however, is the physical atomic diagonal at one fixed determinant shift. A small bounded-projective norm for the row mask is not by itself a comparison between these two quantities. TPC–36 already warned that its $\ell^\infty \widehat{\otimes}_\pi \ell^\infty$ norm is a Schur-multiplier complexity, not a Gram estimate [5].

This paper isolates the finite-dimensional core of that mismatch. It has two logically independent parts.

First, once a coefficient face has genuinely been Hilbertized, the optimal rank-one envelope is not mysterious: it is exactly the nuclear norm. The atomic diagonal is the squared Frobenius norm. Their squared ratio

$$r_{\text{nuc}}(T) := \frac{\|T\|_{S_1}^2}{\|T\|_{S_2}^2} \quad (2)$$

is the correct effective-rank invariant. Algebraic rank gives a valid upper bound, but can be very crude. Stable rank gives no upper bound at all. Moreover, a fixed signed representation may be arbitrarily worse than the optimal nuclear decomposition, because large cancelling layer pairs do not change the coefficient tensor. Thus every fixed-envelope theorem needs two inputs: intrinsic nuclear effective rank and a representation-efficiency estimate.

Second, the bounded projective norm of TPC–36 does become useful after one measures how much weighted mass survives the mask. For $A = D_a K D_b$, set

$$\delta_w(A \mid K) := \frac{\|A\|_{S_2}^2}{\|a\|_2^2 \|b\|_2^2}. \quad (3)$$

We prove the sharp transfer

$$r_{\text{nuc}}(A) \leq \frac{\|K\|_{\pi, \infty}^2}{\delta_w(A \mid K)}. \quad (4)$$

The factor δ_w^{-1} is unavoidable in the class controlled only by bounded projective cost and admitted density: a permutation matrix has projective cost one and attains equality.

The complete-universe extension of the static TPC mask formula has considerably more structure. On the complete dyadic row universe, the three deleted sets—same source prime, near source product, and large row gcd—have uniformly vanishing relative degree. The mask is therefore $1 - o(1)$ dense. A specified explicit TPC–36 representation has soft projective mass, so this complete-universe formula model has only $X^{o(1)}$ projective-to-atomic distortion. This is a positive conclusion: the complete static mask does not account for a remaining polynomial loss.

There is an equally important negative conclusion. The same dense admitted graph has a perfect matching. A permutation packet supported only on that matching has projective cost

one and nuclear effective rank equal to the full row count. It obeys the actual mask support but does not equal the literal joint multiplier. Consequently neither support nor bounded projective complexity can replace a coefficient-specific weighted-density or frame theorem.

1.1 Main results

The paper proves the following statements.

- (i) The Hilbert projective infimum equals the nuclear norm, and the exact energy distortion is r_{nuc} ; see [theorem 3.2](#).
- (ii) For every fixed representation $\mathcal{R}$, the distortion factors exactly as

$$\frac{\mathcal{E}_{\mathcal{R}}(T)^2}{\|T\|_{S_2}^2} = c_{\mathcal{R}}(T)^2 r_{\text{nuc}}(T). \quad (5)$$

No bound uniform over all exact signed representations is possible.

- (iii) Bounded projective cost transfers to weighted atomic energy by [\(4\)](#), for both optimal and specified representations; see [theorem 4.1](#).
- (iv) On a complete dyadic TPC row cell, the complete-universe extension of the literal static-mask formula has minimum degree $(1 - o(1))N$ and specified-representation distortion $X^{o(1)}$; see [theorems 5.1](#) and [5.3](#).
- (v) The actual-mask support contains a sharp permutation packet, while the completed product kernel of TPC-36 gives the same obstruction at orbit-scale modulus. Ferrers separation costs only $\asymp (\log n)^2$ in energy.
- (vi) If the entire isolated-face endpoint allowance is assigned here and $c_{\mathcal{R}} = X^{\chi+o(1)}$ and $r_{\text{nuc}} = X^{\rho+o(1)}$, the exact face-budget condition is $2\chi + \rho \leq 1/400$.

1.2 What is not identified

The nuclear identity applies to a finite Hilbert tensor. The full TPC-48 quantity $\mathfrak{S}_{\mathcal{R}}$ also contains a $C_{\text{LS}}^{1/2}$ -weighted profile norm and an $L^1(d|\nu|)$ reassembly. We do not declare that Banach profile envelope to be a nuclear norm. Nor do we replace its completed coherent prime square by the fixed- h_0 physical diagonal. Those operations would conflate the present mask theorem with prime cancellation, shift localization, and residual grouping.

The result is therefore a genuine but scoped advance. It supplies the correct invariant and closes the complete static-mask model at soft cost; it also proves why the remaining coefficient-specific input cannot be inferred from support alone. It does not prove a fixed-shift Chowla estimate, a parity-barrier breach, or a prime-pair asymptotic.

1.3 Organization

[Section 2](#) records the inherited coefficient interface. [Section 3](#) proves the nuclear identity and the fixed-representation ledger. [Section 4](#) establishes the weighted mask theorem. [Section 5](#) proves the complete-row degree theorem and the matching boundary. [Section 6](#) computes sharp model families. [Section 7](#) inserts the TPC exponents, and [sections 8](#) and [9](#) state the next gate and the deterministic certificate.

2 The inherited coefficient interface

We first state exactly which TPC objects motivate the finite-dimensional theory. This section imports definitions, not an atomic comparison.

2.1 Rows and the literal static mask

On one source-compatible dyadic packet, a row is

$$\alpha = (\ell_\alpha, d_\alpha), \quad m_\alpha = \ell_\alpha d_\alpha \asymp Q, \quad Q = LD, \quad (6)$$

where $\ell_\alpha \asymp L$ is prime and $d_\alpha \asymp D$ is squarefree. The inherited separation makes m_α determine the row on the active packet. TPC-36 defines

$$\mathbf{m}_{\text{act}}(\alpha, \gamma) = \mathbf{1}_{\ell_\alpha \neq \ell_\gamma} \mathbf{1}_{|m_\alpha - m_\gamma| > \Delta} \mathbf{1}_{(d_\alpha, d_\gamma) \leq G}, \quad (7)$$

with

$$\Delta = QX^{-\kappa_0}, \quad G = X^{\kappa_0}, \quad \kappa_0 > 0. \quad (8)$$

The literal joint multiplier is

$$\mathfrak{A}_{\alpha, \gamma}^{\text{act}}(j) = \mathbf{m}_{\text{act}}(\alpha, \gamma) \Xi_{\alpha, \gamma}(j) \mathcal{W}_{\alpha, \gamma}(j/J). \quad (9)$$

Here Ξ is fixed-period in j and $\mathcal{W}$ contains smooth cutoffs. After splitting the fixed residue classes and applying the inherited Mellin separation, the full multiplier has an exact signed projective representation with every fixed orbit seminorm having $X^{o(1)}$ total variation [5].

For a finite kernel $K : \mathcal{I} \times \mathcal{J} \rightarrow \mathbb{C}$, the relevant TPC-36 norm is

$$\|K\|_{\pi, \infty} := \inf_{K = \int p_\omega q_\omega d\nu(\omega)} \int \|p_\omega\|_\infty \|q_\omega\|_\infty d|\nu|(\omega). \quad (10)$$

The exact static-mask theorem gives

$$\|\mathbf{m}_{\text{act}}\|_{\pi, \infty} \ll \log(2Q)(3 + 2\sqrt{2})^{\omega_*} = X^{o(1)}, \quad (11)$$

where ω_* is the maximum number of prime factors of an active row divisor. This is an unconditional identity and upper bound on the declared packet. It is not a nuclear-norm or Gram estimate.

2.2 The physical atomic diagonal

TPC-45 writes the literal stripped equality atoms as $b_{i,u}$, where i records an output row and an orbit time and u is the squarefree equality label. Its atomic diagonal is

$$\mathcal{D}_0 := \sum_i \sum_u |b_{i,u}|^2. \quad (12)$$

At the all-minus prime-sign point the associated Walsh polynomial recovers the literal physical equality columns [6]. The quantity $\mathcal{D}_0$ is therefore tied to one fixed physical shift and to the native atomic indexing. It is not the Cartesian energy of an arbitrary completed determinant family.

After large-prime descent, the left residual synthesis at the physical shift h_0 has the form

$$\mathcal{F}_{h_0}^{\text{act}, \text{L}} = \sum_{\substack{\gamma, j, m, p, r \\ pr - mj = h_0}} \gamma_m^{(1)} \mathfrak{A}_{m, \gamma}^{\text{act}}(j) \mu(r) \Lambda(pr) e_{\gamma, d(m)r} \otimes e_j. \quad (13)$$

It uses literal coefficients, but belongs to the low-sign-averaged residual Gram route rather than the native affine physical face [6, 7].

2.3 The frozen projective row-square

TPC-47/48 freezes every coefficient constructed at h_0 and replaces the determinant equation by $pr - mj = h$ for arbitrary $h \in \mathbb{Z}$. Only $h = h_0$ has physical provenance. One exact projective layer is

$$\mathcal{F}_h^\omega = \sum_{p,r,m} \sum_{j \equiv a_\omega \pmod{H_0}} A_\omega(p, r, m) \phi_\omega(j/J) v_{\omega,p,r,m} \otimes e_j \mathbf{1}_{pr-mj=h}. \quad (14)$$

The complete frozen family is the signed Bochner integral of all such layers. Fourier transformation in h defines

$$B_{\omega,m}(\alpha) := \sum_{p,r} A_\omega(p, r, m) e(-pr\alpha) v_{\omega,p,r,m}. \quad (15)$$

At fixed m , distinct r have orthogonal residual labels, but the inner p -sum remains coherent. For a fixed admissible global representation $\mathcal{R}$, TPC-48 introduces

$$\mathfrak{S}_{\mathcal{R}}(\alpha) := \int C_{\text{LS}}(\phi_\omega, H_0)^{1/2} \left(\sum_m \|B_{\omega,m}(\alpha)\|^2 \right)^{1/2} d|\nu|(\omega). \quad (16)$$

Its pointwise infimum over admissible representations is $\mathfrak{S}_{\text{pr}}$. The infimum is not integrated; the annular theorem retains one fixed measurable global representation [8].

Three differences from a finite matrix nuclear norm must remain visible:

- (a) $C_{\text{LS}}^{1/2}$ is a profile seminorm weight, not a declared Hilbert norm;
- (b) the p -sum in (15) is taken before the square; and
- (c) $\mathfrak{S}_{\mathcal{R}}(\alpha)$ belongs to a completed frozen field, whereas $\mathcal{D}_0$ belongs to one physical slice.

The finite theory below applies exactly to Hilbertized coefficient faces and supplies a criterion for each separated mask cell. A full comparison of (16) with (12) needs additional theorems and is not silently imported.

3 Hilbert projective geometry and the exact distortion

Let $\mathcal{H}_1, \mathcal{H}_2$ be finite-dimensional Hilbert spaces and identify $T \in \mathcal{H}_1 \otimes \overline{\mathcal{H}_2}$ with an operator $\mathcal{H}_2 \rightarrow \mathcal{H}_1$. Standard facts about projective tensor products, nuclear operators, and Hilbert–Schmidt operators may be found in [1, 3, 4]; the short proofs needed here are included.

Definition 3.1 (Hilbert projective envelope). For a rank-one representation written as

$$\mathcal{R} : \quad T = \sum_{\omega \in \Omega} x_\omega \otimes \overline{y_\omega}, \quad (17)$$

define its envelope and optimum by

$$\mathcal{E}_{\mathcal{R}}(T) := \sum_{\omega \in \Omega} \|x_\omega\|_2 \|y_\omega\|_2, \quad \mathcal{E}_{\text{pr}}(T) := \inf_{\mathcal{R}} \mathcal{E}_{\mathcal{R}}(T). \quad (18)$$

Finite sums suffice. A Bochner integral gives the same infimum after finite-dimensional approximation.

Theorem 3.2 (Hilbert projective equals nuclear). *If $T \neq 0$ has singular values $\sigma_1, \dots, \sigma_r > 0$, then*

$$\mathcal{E}_{\text{pr}}(T) = \|T\|_{S_1} = \sum_{k=1}^r \sigma_k, \quad (19)$$

$$\mathcal{D}_{\text{at}}(T) := \|T\|_{S_2}^2 = \sum_{k=1}^r \sigma_k^2, \quad (20)$$

$$\mathcal{E}_{\text{pr}}(T)^2 = r_{\text{nuc}}(T) \mathcal{D}_{\text{at}}(T), \quad r_{\text{nuc}}(T) := \frac{(\sum_k \sigma_k)^2}{\sum_k \sigma_k^2}. \quad (21)$$

Moreover

$$1 \leq \text{sr}(T) := \frac{\|T\|_{S_2}^2}{\|T\|_{\text{op}}^2} \leq r_{\text{nuc}}(T) \leq \text{rank } T. \quad (22)$$

The upper equality holds exactly when all nonzero singular values are equal; the lower endpoint $r_{\text{nuc}} = 1$ holds exactly at rank one.

Proof. The singular-value decomposition $T = \sum_k \sigma_k u_k \otimes \overline{v_k}$ gives $\mathcal{E}_{\text{pr}}(T) \leq \sum_k \sigma_k$. Conversely, the nuclear norm is a norm and every rank-one operator satisfies $\|x \otimes \overline{y}\|_{S_1} = \|x\|_2 \|y\|_2$. Thus every representation (17) has $\|T\|_{S_1} \leq \mathcal{E}_{\mathcal{R}}(T)$, proving (19). Formula (20) is the Hilbert–Schmidt identity.

Cauchy–Schwarz gives $(\sum_k \sigma_k)^2 \leq r \sum_k \sigma_k^2$, with equality exactly for equal positive singular values. Also $\sigma_1 \sum_k \sigma_k \geq \sum_k \sigma_k^2$, whence

$$r_{\text{nuc}}(T) \geq \frac{\sum_k \sigma_k^2}{\sigma_1^2} = \text{sr}(T). \quad (23)$$

The remaining equality statements are immediate. $\square$

Remark 3.3 (Stable rank points in the wrong direction). Stable rank is a lower, not an upper, bound for the present distortion. Let T_N have singular values 1 and $N^{-3/4}$, the latter repeated N times. Then

$$\text{sr}(T_N) = 1 + N^{-1/2} \longrightarrow 1, \quad r_{\text{nuc}}(T_N) = \frac{(1 + N^{1/4})^2}{1 + N^{-1/2}} \asymp N^{1/2}. \quad (24)$$

Thus the nuclear participation rank, not ordinary stable rank, is the correct parameter.

3.1 A fixed representation has a second condition number

Definition 3.4 (Representation cancellation number). For a fixed exact representation $\mathcal{R}$ of $T \neq 0$, define

$$c_{\mathcal{R}}(T) := \frac{\mathcal{E}_{\mathcal{R}}(T)}{\|T\|_{S_1}} \geq 1. \quad (25)$$

Proposition 3.5 (Exact fixed-representation ledger). *For every fixed exact rank-one representation,*

$$\boxed{\frac{\mathcal{E}_{\mathcal{R}}(T)^2}{\mathcal{D}_{\text{at}}(T)} = c_{\mathcal{R}}(T)^2 r_{\text{nuc}}(T)}. \quad (26)$$

There is no finite upper bound for the left side that is uniform over all exact signed representations of one fixed nonzero tensor.

Proof. The identity is the product of the definitions. For nonuniformity, fix a nonzero rank-one tensor $z = x \otimes \bar{y}$, normalized so that $\|x\|\|y\| = 1$. For every $R > 0$, the identity

$$z = (R + 1)z - Rz \quad (27)$$

has envelope $2R + 1$, although the represented tensor, its rank, and its atomic energy remain unchanged. More generally, the null pair $Rz - Rz$ can be appended to any representation. $\square$

This elementary obstruction is consequential for (16). A theorem for a specified TPC representation may estimate its cancellation number. A pointwise infimum may suppress artificial null pairs. A claim for every exact signed representation cannot be true.

3.2 Two usable positive criteria

Corollary 3.6 (Approximate nuclear rank). *Suppose the singular values are decreasing and, for some $R \geq 1$,*

$$\sum_{k>R} \sigma_k \leq \eta \|T\|_{S_2}. \quad (28)$$

Then

$$\|T\|_{S_1} \leq (\sqrt{R} + \eta) \|T\|_{S_2}, \quad r_{\text{nuc}}(T) \leq (\sqrt{R} + \eta)^2. \quad (29)$$

Proof. Cauchy–Schwarz bounds the first R singular values by $\sqrt{R}\|T\|_{S_2}$; add (28). $\square$

Proposition 3.7 (Orthogonal layers do not improve an L^1 envelope). *Suppose the rank-one tensors $z_1, \dots, z_s$ are pairwise orthogonal in Hilbert–Schmidt inner product, put $a_k = \|z_k\|_{S_2}$, and use the displayed decomposition $T = \sum_k z_k$. Then*

$$\mathcal{D}_{\text{at}}(T) = \sum_k a_k^2, \quad \mathcal{E}_{\mathcal{R}}(T)^2 = N_{\text{eff}}(a) \mathcal{D}_{\text{at}}(T), \quad N_{\text{eff}}(a) := \frac{(\sum_k a_k)^2}{\sum_k a_k^2}. \quad (30)$$

Equal layers lose the full layer count s .

Thus orthogonality helps only if it is retained through an L^2 /Bessel reassembly. Applying Minkowski first converts it into an L^1 layer count. Entrywise positivity or one-sign coefficients do not repair this: permutation matrices below are nonnegative and still attain maximal nuclear distortion.

4 A sharp weighted transfer from bounded projective masks

Let I, J be finite sets, let $K : I \times J \rightarrow \mathbb{C}$, and let $a = (a_i)_{i \in I}$, $b = (b_j)_{j \in J}$ be nonnegative weights. Zero weights may be deleted. Write

$$A := D_a K D_b, \quad A_{ij} = a_i K_{ij} b_j. \quad (31)$$

Complex unary row and column factors cause no loss of generality here: their moduli are placed in a, b , while their unit phases are absorbed into K . Such unit-modulus row and column Schur factors preserve every bounded-projective cost used below. When $A \neq 0$, define its weighted survival parameter by

$$\delta_w(K; a, b) := \frac{\|A\|_{S_2}^2}{\|a\|_2^2 \|b\|_2^2} = \frac{\sum_{i,j} a_i^2 |K_{ij}|^2 b_j^2}{(\sum_i a_i^2) (\sum_j b_j^2)}. \quad (32)$$

For a 0-1 mask it lies in $[0, 1]$ and is literally the proportion of product ℓ^2 -mass lying on admitted pairs.

Fix an exact bounded-projective representation

$$\mathcal{R} : \quad K_{ij} = \int_{\Omega} p_{\omega}(i) q_{\omega}(j) d\nu(\omega), \quad (33)$$

and put

$$P_{\mathcal{R}}(K) := \int_{\Omega} \|p_{\omega}\|_{\infty} \|q_{\omega}\|_{\infty} d|\nu|(\omega), \quad (34)$$

$$\mathcal{E}_{\mathcal{R}}(K; a, b) := \int_{\Omega} \|D_a p_{\omega}\|_2 \|D_b \overline{q_{\omega}}\|_2 d|\nu|(\omega). \quad (35)$$

The conjugation merely matches the operator convention and has no effect on any norm.

Theorem 4.1 (Weighted projective-to-atomic transfer). *For every representation (33),*

$$\|A\|_{S_1} \leq \mathcal{E}_{\mathcal{R}}(K; a, b) \leq P_{\mathcal{R}}(K) \|a\|_2 \|b\|_2, \quad (36)$$

$$\frac{\mathcal{E}_{\mathcal{R}}(K; a, b)^2}{\|A\|_{S_2}^2} \leq \frac{P_{\mathcal{R}}(K)^2}{\delta_w(K; a, b)}. \quad (37)$$

Taking the infimum in the bounded projective cost gives

$$r_{\text{nuc}}(A) \leq \min \left\{ \text{rank } A, \frac{\|K\|_{\pi, \infty}^2}{\delta_w(K; a, b)} \right\}. \quad (38)$$

If $a_i, b_j > 0$ on the active sets and one takes the infimum of (35) over all exact representations there, the result is exactly $\|A\|_{S_1}$. This unrestricted infimum is not asserted to coincide with the infimum over the narrower inherited admissible TPC-48 representation class.

Proof. After multiplying (33) by the two diagonal weights,

$$A = \int_{\Omega} (D_a p_{\omega}) \otimes (\overline{D_b q_{\omega}}) d\nu(\omega). \quad (39)$$

The nuclear norm is at most the integral of the rank-one nuclear norms, which is the first inequality in (36). Pointwise,

$$\|D_a p_{\omega}\|_2 \|D_b \overline{q_{\omega}}\|_2 \leq \|a\|_2 \|b\|_2 \|p_{\omega}\|_{\infty} \|q_{\omega}\|_{\infty}. \quad (40)$$

Integration proves the second inequality. Divide its square by $\|A\|_{S_2}^2 = \delta_w \|a\|_2^2 \|b\|_2^2$ to obtain (37). Infimize and combine with theorem 3.2 for (38). Finally, when the weights are positive, every rank-one decomposition of A can be divided by a_i, b_j to give a representation of K on the active rectangle; the exact infimum is therefore the nuclear norm. $\square$

Remark 4.2 (Sufficient and exact criteria). The exact intrinsic criterion is $r_{\text{nuc}}(A)$, and for a fixed representation it is $c_{\mathcal{R}}(A)^2 r_{\text{nuc}}(A)$. The quantity $\|K\|_{\pi, \infty}^2 / \delta_w$ is a computable sufficient upper bound. A lower bound on δ_w is not necessary for a particular matrix: a sparse rank-one rectangle can have tiny density and $r_{\text{nuc}} = 1$. The density dependence is instead sharp in the minimax class controlled only by bounded projective cost and δ_w , as proposition 4.4 shows.

4.1 A uniform degree criterion

For a 0-1 mask K with $|J| = N$, suppose every row has at most ϵN forbidden entries. Define the flatness of the right weight by

$$\text{Flat}(b) := \frac{N \|b\|_{\infty}^2}{\|b\|_2^2} \geq 1. \quad (41)$$

Corollary 4.3 (Degree plus flatness). *For arbitrary nonzero left weights a ,*

$$\delta_w(K; a, b) \geq 1 - \epsilon \text{Flat}(b). \quad (42)$$

Consequently, if the right side is positive,

$$r_{\text{nuc}}(D_a K D_b) \leq \frac{\|K\|_{\pi, \infty}^2}{1 - \epsilon \text{Flat}(b)}. \quad (43)$$

The same statement holds with the two sides exchanged.

Proof. For each row i , its forbidden b^2 -mass is at most $\epsilon N \|b\|_\infty^2 = \epsilon \text{Flat}(b) \|b\|_2^2$. Average the complementary inequality with weights a_i^2 , then use [theorem 4.1](#). $\square$

This criterion tolerates arbitrary concentration in the source weight provided the opposite-row weight is flat. For the TPC joint multiplier, flatness and admitted density of the active smooth/residue cell are new conditions: the inherited interface gives upper seminorms and exact identities, but no lower-density theorem.

4.2 The sharp permutation boundary

Proposition 4.4 (Permutation equality). *Let P_n be any $n \times n$ permutation matrix and take $a = b = \mathbf{1}$. Then*

$$\|P_n\|_{\pi, \infty} = 1, \quad \delta_w(P_n; \mathbf{1}, \mathbf{1}) = \frac{1}{n}, \quad \|P_n\|_{S_1} = n, \quad \|P_n\|_{S_2}^2 = n. \quad (44)$$

Thus $r_{\text{nuc}}(P_n) = n$, and (38) is an equality.

Proof. Permutations preserve singular values, so all n singular values are one. For the identity matrix,

$$I_n = 2^{-n} \sum_{\epsilon \in \{\pm 1\}^n} \epsilon \epsilon^\top. \quad (45)$$

Every factor has supremum norm one, giving bounded projective cost at most one; the maximum entry gives the reverse inequality. Permuting a factor preserves the cost. The remaining formulas follow by counting the n nonzero entries. $\square$

The example is nonnegative, one-sign, and has an orthogonal singular basis. None of those qualitative properties replaces a quantitative effective-rank or density estimate.

5 The complete-universe extension of the static-mask formula

The inherited active TPC packet may be pruned by smooth and residue weights, and no lower row count is part of the TPC-45/47 interface. We therefore separate two statements. This section first analyzes the complete-universe extension on which the literal static-mask formula (7) is defined. It then states the weighted condition needed to transfer the conclusion to a pruned active cell.

5.1 The ambient row universe

Fix $H \geq 1$ and $\kappa_0 > 0$. Let $L, D \rightarrow \infty$ with X , assume $L, D \leq X^C$ for some fixed C , and let

$$\mathcal{R}(L, D; H) := \{(\ell, d) : L < \ell \leq 2L, \ell \in \mathbb{P}, D < d \leq 2D, \mu^2(d) = 1, (d, H) = 1\}. \quad (46)$$

Assume $L > 4D$, as is more than satisfied by the inherited endpoint separation. Put $N = |\mathcal{R}(L, D; H)|$. The prime number theorem and the elementary squarefree count in fixed coprime residue conditions give

$$N \asymp_H \frac{LD}{\log L} = X^{o(1)} LD. \quad (47)$$

Only this two-sided order estimate is used; see, for example, [2, Chapter 1]. The row product $m = \ell d$ is injective: if $\ell d = \ell' d'$ and $\ell \neq \ell'$, then the prime ℓ divides $d' < 2D < \ell$, a contradiction.

Use the same row universe on the two sides and define $M_X(\alpha, \gamma)$ by the literal static formula (7), with

$$Q = LD, \quad \Delta = QX^{-\kappa_0}, \quad G = X^{\kappa_0}. \quad (48)$$

Theorem 5.1 (Uniform degree of the complete static mask). *Uniformly in every row on either side,*

$$\deg_{M_X}(\alpha) \geq (1 - \epsilon_X)N, \quad \epsilon_X \ll_H L^{-1+o(1)} + D^{-1+o(1)} + X^{-\kappa_0+o(1)}. \quad (49)$$

In particular $\epsilon_X = o(1)$ whenever $L, D \rightarrow \infty$.

Proof. Fix $\alpha = (\ell, d)$. The forbidden rows lie in the union of three sets.

For the same-source deletion $\ell_\gamma = \ell$, there are $O_H(D)$ choices. Dividing by (47) gives

$$\frac{O_H(D)}{N} \ll_H \frac{\log L}{L} = L^{-1+o(1)}. \quad (50)$$

For the near-row deletion, injectivity of the integer label m_γ gives at most $2\Delta + 1$ rows. Hence

$$\frac{2\Delta + 1}{N} \ll_H X^{-\kappa_0} \log L + \frac{\log L}{LD} = X^{-\kappa_0+o(1)} + (LD)^{-1+o(1)}. \quad (51)$$

Finally, if $(d, d_\gamma) > G$, the gcd is a divisor $g \mid d$ with $g > G$. For each opposite source prime, the number of possible d_γ is at most

$$\sum_{\substack{g \mid d \\ g > G}} \left(\frac{2D}{g} + 1 \right) \leq \tau(d) \left(\frac{2D}{G} + 1 \right). \quad (52)$$

Here we used $g^{-1} \leq G^{-1}$ and at most $\tau(d)$ divisors. Since $\tau(d) = D^{o(1)} = X^{o(1)}$ under the declared polynomial scale bound, multiplication by the $O(L/\log L)$ source primes and division by N yield

$$\frac{\#\{\gamma : (d, d_\gamma) > G\}}{N} \ll_H X^{-\kappa_0+o(1)} + D^{-1+o(1)}. \quad (53)$$

The union bound proves (49). The formula is symmetric, so the column estimate is identical. $\square$

Corollary 5.2 (Density and a perfect matching). *The number of admitted entries is*

$$\|M_X\|_{S_2}^2 = \#\{(\alpha, \gamma) : M_X(\alpha, \gamma) = 1\} = (1 + O(\epsilon_X))N^2. \quad (54)$$

For all sufficiently large X , the admitted bipartite graph has a perfect matching.

Proof. Sum the row degrees for the first statement. For large X , both the minimum row degree and minimum column degree exceed $N/2$. The empty set is trivial. If $0 < |S| \leq N/2$, any vertex in S has more than $N/2$ neighbors, so $|\Gamma(S)| > |S|$. If $|S| > N/2$, a right vertex without a neighbor in S would have all its more than $N/2$ neighbors in the complement of S , whose size is less than $N/2$, a contradiction. Hall's condition follows. $\square$

5.2 A specified explicit complete-mask comparison

Fix one explicit representation $\mathcal{R}_{36}$ furnished by the proof of the TPC-36 static-mask theorem, using same-source orthogonality, signed Ferrers separation, and global gcd compression. That construction, rather than the infimum alone in (11), gives a fixed-representation bound. Define

$$\omega_*^{\text{comp}} := \max_{\substack{D < d \leq 2D \\ \mu^2(d)=1, (d,H)=1}} \omega(d), \quad (55)$$

where $\omega(d)$ counts the distinct prime divisors of d . Then

$$P_{\mathcal{R}_{36}}(M_X) \ll \log(2Q)(3 + 2\sqrt{2})\omega_*^{\text{comp}} = X^{o(1)}. \quad (56)$$

Indeed, every fixed-base exponential in ω_*^{comp} is $D^{o(1)} = X^{o(1)}$.

Theorem 5.3 (Soft atomic cost for the complete-universe extension). *For all sufficiently large X , on $\mathcal{R}(L, D; H) \times \mathcal{R}(L, D; H)$,*

$$\frac{\mathcal{E}_{\mathcal{R}_{36}}(M_X; \mathbf{1}, \mathbf{1})^2}{\|M_X\|_{S_2}^2} \leq \frac{P_{\mathcal{R}_{36}}(M_X)^2}{1 - \epsilon_X} = X^{o(1)}. \quad (57)$$

Consequently

$$r_{\text{nuc}}(M_X) = X^{o(1)}. \quad (58)$$

Proof. For uniform weights, the minimum-degree bound gives $\delta_w \geq 1 - \epsilon_X$. Apply theorem 4.1 to the specified explicit representation and then use (56). The intrinsic infimum is no larger. $\square$

This theorem concerns the complete static 0-1 mask. It does not say that every coefficient supported by the mask has soft nuclear rank, nor that the full multiplier (9) has a weighted lower density on every active residue/Mellin cell.

Corollary 5.4 (A criterion for a pruned weighted cell). *Suppose a retained opposite-row weight b on the complete universe satisfies*

$$\epsilon_X \text{Flat}(b) = o(1). \quad (59)$$

Then for every nonzero source weight a ,

$$\frac{\mathcal{E}_{\mathcal{R}_{36}}(M_X; a, b)^2}{\|D_a M_X D_b\|_{S_2}^2} \leq X^{o(1)}. \quad (60)$$

This is a checkable sufficient condition for a separated TPC cell. The current inherited interface does not supply (59); it is a new active-cell task.

5.3 A support-class obstruction inside the same mask

Let σ be the perfect matching from corollary 5.2 and define

$$P_\sigma(\alpha, \gamma) = \mathbf{1}_{\gamma=\sigma(\alpha)}. \quad (61)$$

Then P_σ is supported entirely where $M_X = 1$.

Theorem 5.5 (Sharp actual-mask-support witness). *The packet (61) satisfies*

$$\|P_\sigma\|_{\pi, \infty} = 1, \quad \|P_\sigma\|_{S_1} = N, \quad \|P_\sigma\|_{S_2}^2 = N, \quad r_{\text{nuc}}(P_\sigma) = N. \quad (62)$$

At the inherited endpoint,

$$N = X^{267/400+o(1)}. \quad (63)$$

Thus actual-mask support plus bounded projective cost $X^{o(1)}$ cannot control arbitrary supported coefficients at atomic scale.

Proof. The matrix is a permutation, so [proposition 4.4](#) applies. The endpoint row count is $N = X^{o(1)}LD = X^{267/400+o(1)}$. $\square$

The witness deliberately zeros all other admitted entries. It is not the literal multiplier $\gamma_m \mathfrak{A}_{m,\gamma}^{\text{act}}(j)$, and no claim is made that the physical prime/cofactor packet realizes it. Constructing a polynomial family of literal matching atoms would require arithmetic information about $mj + h_0$ and its eligible large prime factor; that information is precisely outside the present gate.

6 Sharp model families and false shortcuts

The weighted theorem contains two independent inputs: bounded projective cost and admitted density. The following exact families show why both matter and why several qualitative substitutes fail.

6.1 Ferrers masks spend only logarithmic energy

Let

$$T_n = (\mathbf{1}_{k \leq r})_{1 \leq r, k \leq n} \quad (64)$$

be the lower-triangular summation matrix.

Proposition 6.1 (Exact Ferrers distortion). *The singular values of T_n are*

$$\sigma_k(T_n) = \left[2 \sin \frac{(2k-1)\pi}{4n+2} \right]^{-1}, \quad 1 \leq k \leq n. \quad (65)$$

Moreover

$$\|T_n\|_{S_2}^2 = \frac{n(n+1)}{2}, \quad (66)$$

$$\|T_n\|_{S_1} = \frac{n}{\pi} \log n + O(n), \quad (67)$$

$$r_{\text{nuc}}(T_n) = \frac{2}{\pi^2} (\log n)^2 + O(\log n). \quad (68)$$

Proof. The inverse of T_n is the lower bidiagonal first-difference matrix. Diagonalizing $(T_n^{-1})^* T_n^{-1}$ by the discrete sine basis gives (65); this is also the exact calculation used in TPC-36. Counting ones gives (66). For $k \leq n/2$, where the corresponding angle satisfies $0 < \theta_k \leq \pi/4$, use uniformly

$$\frac{1}{2 \sin \theta} = \frac{1}{2\theta} + O(\theta) \quad (69)$$

and $\sum_{k \leq n/2} (2k-1)^{-1} = \frac{1}{2} \log n + O(1)$. The remaining singular values sum to $O(n)$. This proves (67); division by (66) gives the last formula. $\square$

TPC-36 proves $\|T_n\|_{\pi, \infty} \asymp \log n$, while its unweighted admitted density tends to $1/2$. Thus [theorem 4.1](#) has the correct order for this family. At every polynomial $n = X^{O(1)}$, the energy cost is $X^{o(1)}$. Signed Ferrers separation is not a polynomial Door-B obstruction.

6.2 Product kernels and product phases

Let $q > 2$ be prime and write $G_q = \mathbb{F}_q^\times$. For $a \in G_q$, define

$$K_a(x, y) = \mathbf{1}_{xy=a}, \quad x, y \in G_q. \quad (70)$$

Proposition 6.2 (Cost-one product permutation). *The matrix K_a is a permutation of size $q-1$, and*

$$\|K_a\|_{\pi, \infty} = 1, \quad r_{\text{nuc}}(K_a) = q-1. \quad (71)$$

Proof. For each x , the equation $xy = a$ has the unique solution $y = ax^{-1}$. Apply [proposition 4.4](#). Equivalently, multiplicative-character orthogonality gives a cost-one projective representation. $\square$

This is the same product-delta kernel that appears in the completion analysis of TPC-36. It is a method-specific finite-field obstruction, not the literal residual coefficient. At the endpoint one may choose a prime $q \asymp J$ by Bertrand's postulate, so the model dimension has precisely the orbit-scale exponent.

For $r \neq 0$, also define the flat product phase

$$A_r(x, y) = e_q(rxy), \quad e_q(z) = \exp(2\pi iz/q). \quad (72)$$

Proposition 6.3 (Flat entries do not imply low nuclear rank). *The singular values of A_r are $\sqrt{q}$, with multiplicity $q - 2$, and 1, with multiplicity one. Therefore*

$$\|A_r\|_{S_2}^2 = (q - 1)^2, \quad (73)$$

$$r_{\text{nuc}}(A_r) = \left(\frac{(q - 2)\sqrt{q} + 1}{q - 1} \right)^2 \asymp q. \quad (74)$$

The last displayed square is exactly the square of the bounded projective norm computed in TPC-36.

Proof. Additive orthogonality gives $A_r A_r^* = qI_{q-1} - J_{q-1}$. Its eigenvalues are q on the orthogonal complement of constants and 1 on constants. The stated Schatten norms follow. Multiplicative characters give flat singular vectors and attain the bounded projective lower bound $\|A_r\|_{S_1}/(q-1)$; see [\[5\]](#). $\square$

Thus flat entries, flat singular vectors, and the usual upper-bound notion of incoherence do not force a small projective-to-atomic loss. They do not control the number of comparable singular values.

6.3 Positive density alone also fails

The permutation example has small density. A dense nonnegative example shows that density without bounded projective cost is equally insufficient. Let $n = 4^k$, put

$$H_4 = J_4 - 2I_4, \quad H_n = H_4^{\otimes k}, \quad K_n = \frac{J_n + H_n}{2}. \quad (75)$$

Then H_n has entries ± 1 , satisfies $H_n H_n^* = nI_n$, and has constant row sum $\sqrt{n}$. Hence K_n is a 0-1 matrix.

Proposition 6.4 (Dense positive high-rank mask). *The singular values of K_n are*

$$\frac{n + \sqrt{n}}{2} \quad \text{once,} \quad \frac{\sqrt{n}}{2} \quad \text{with multiplicity } n - 1. \quad (76)$$

Consequently

$$\delta_w(K_n; \mathbf{1}, \mathbf{1}) = \frac{1 + n^{-1/2}}{2}, \quad r_{\text{nuc}}(K_n) = \frac{n + \sqrt{n}}{2}. \quad (77)$$

In particular, $\|K_n\|_{\pi, \infty} \geq (1 + \sqrt{n})/2$.

Proof. Using $J_n H_n = H_n J_n = \sqrt{n} J_n$,

$$K_n K_n^* = \frac{1}{4} ((n + 2\sqrt{n})J_n + nI_n). \quad (78)$$

This gives (76). The number of ones is $\|K_n\|_{S_2}^2 = n(n + \sqrt{n})/2$, while $\|K_n\|_{S_1} = n(1 + \sqrt{n})/2$; division proves (77). Finally $\|K_n\|_{\pi, \infty} \geq \|K_n\|_{S_1}/n$ by the elementary nuclear lower bound for the bounded projective norm. $\square$

The two hypotheses in [theorem 4.1](#) are therefore genuinely complementary: projective cost one with small density can be bad, and positive density with polynomial projective cost can be bad. The complete TPC static mask is favorable because it has both soft cost and density $1 - o(1)$.

7 Endpoint ledger and the isolated-face Door-B threshold

Write the inherited exponents as

$$\mu := \frac{267}{400}, \quad \nu := \frac{133}{400}, \quad \kappa := \frac{1}{400}, \quad \mu + \nu = 1. \quad (79)$$

Thus the source-row scale is $Q = X^{\mu+o(1)}$, the orbit scale is $J = X^{\nu+o(1)}$, and the inherited determinant-packet energy allowance is $X^{\kappa+o(1)}$ [\[6–8\]](#).

7.1 The exact isolated-face budget criterion

Suppose one finite Hilbertized coefficient face has

$$c_{\mathcal{R}}(T) = X^{\chi+o(1)}, \quad r_{\text{nuc}}(T) = X^{\rho+o(1)}. \quad (80)$$

By [proposition 3.5](#), the fixed representation loses the exact energy exponent

$$2\chi + \rho. \quad (81)$$

If the entire $X^{\kappa+o(1)}$ allowance is assigned to this isolated Hilbertized face, then the face fits that budget exactly when

$$\boxed{2\chi + \rho \leq \frac{1}{400}} \quad (82)$$

up to the declared $o(1)$ convention. This is not a necessary-and-sufficient criterion for full TPC–48 closure. If the remaining coefficient-specific collision, prime-square, grouping, and localization operations consume a combined exponent $\lambda_{\text{other}} \geq 0$, the corresponding multiplicative ledger instead requires

$$2\chi + \rho + \lambda_{\text{other}} \leq \frac{1}{400}. \quad (83)$$

Those other losses are not estimated in this paper.

For an optimal Hilbert projective representation, $\chi = 0$, so the isolated-face criterion is $r_{\text{nuc}} \leq X^{1/400+o(1)}$. For a named bounded-projective representation with

$$P_{\mathcal{R}} = X^{\varpi+o(1)}, \quad \delta_w = X^{-\theta+o(1)}, \quad (84)$$

[theorem 4.1](#) gives the sufficient condition

$$2\varpi + \theta \leq \frac{1}{400}. \quad (85)$$

When $P_{\mathcal{R}} = X^{o(1)}$, the clean target is

$$\delta_w \geq X^{-1/400+o(1)}. \quad (86)$$

This density condition is minimax-sharp by permutations, but is only sufficient for a specified coefficient matrix. The exact criterion for the isolated face remains [\(82\)](#).

7.2 Model costs at the endpoint

The following table uses the energy convention throughout.

face or model	effective energy loss	excess over $X^{1/400}$
complete-universe static-mask formula model	$X^{o(1)}$	none in this model
signed Ferrers mask, $n = X^{O(1)}$	$\asymp (\log n)^2 = X^{o(1)}$	none
product permutation, prime $q \asymp J$	$X^{133/400+o(1)}$	$X^{33/100+o(1)}$
actual-mask-support matching, $N = QX^{o(1)}$	$X^{267/400+o(1)}$	$X^{133/200+o(1)}$
dense-slope common-profile collision at the TPC-48 coarse edge	$X^{67/200+o(1)}$	$X^{133/400+o(1)}$

The first line is the positive theorem (57) for the complete-universe extension of the literal formula. It is not a theorem for an arbitrarily pruned active joint multiplier. The matching line is supported inside the literal static-mask graph but uses artificial coefficient values. The product-permutation line is a completed finite-field model. Neither is a lower bound for the literal prime/cofactor coefficient.

The last line is a different obstruction. TPC-48 proves that common orbit profiles improve unrestricted slope synthesis, yet a dense consecutive-slope model at the coarse edge still has collision factor

$$X^{\mu-\nu+o(1)} = X^{67/200+o(1)}. \quad (87)$$

The present static-mask theorem does not cancel that factor. Nuclear distortion and determinant-slope collision are separate multiplicative ledgers.

7.3 What a full active-cell theorem would have to say

For a separated residue/Mellin cell of the actual joint multiplier, a usable theorem may take either of two forms:

- (a) a weighted admission and specified-representation efficiency estimate satisfying (85); or
- (b) a direct singular-value estimate satisfying (82), possibly by corollary 3.6.

The former can use corollary 4.3 after proving opposite-row flatness of the literal active weight. The latter could exploit coefficient-specific low effective rank even when admitted density is small.

After the individual cells are controlled, one still needs a frame or no-cancellation theorem for their signed $L^1(d|\nu|)$ reassembly. Orthogonality of formal layers is not enough under Minkowski by proposition 3.7. Finally, the coherent prime square inside (15) and the completed-to- h_0 normalization must be treated separately. These are arithmetic and localization inputs, not missing steps in the nuclear calculation.

8 Claim boundary and the next gate

8.1 What is proved

The present paper establishes the following reusable conclusions.

- (i) On every genuinely Hilbertized coefficient face, the optimal projective-to-atomic energy loss is exactly the nuclear participation rank.

- (ii) A fixed signed representation has an independent cancellation condition number; appending null pairs proves that it cannot be omitted.
- (iii) The TPC-36 bounded projective norm transfers to the Hilbert envelope with the sharp weighted-density factor δ_w^{-1} .
- (iv) On the complete dyadic row universe, the complete-universe extension of the literal static-mask formula has density $1 - o(1)$, a perfect matching, and only soft distortion for the specified explicit representation.
- (v) The admitted support nevertheless contains a permutation packet with full row-count distortion. Thus support-only methods and bounded projective complexity alone are sharply insufficient for arbitrary supported coefficients.
- (vi) If the entire endpoint allowance is assigned to the isolated Hilbertized face, its exact ledger is $2\chi + \rho \leq 1/400$, while the computable density route gives the sufficient target $2\varpi + \theta \leq 1/400$.

The static-mask result is not merely an abstract norm restatement. It removes one possible polynomial obstruction on the complete row model and replaces a vague “projective-to-atomic” request by two measurable quantities: nuclear effective rank and representation efficiency.

8.2 What is not proved

None of the following is asserted.

- (i) The inherited active row packet is not assumed to equal the complete universe (46). The current interface gives row-count upper bounds, not weighted lower density or flatness.
- (ii) The support-class matching packet is not the literal physical multiplier, and the product kernel is not the literal residual prime/cofactor coefficient.
- (iii) The full Banach profile envelope $\mathfrak{S}_{\mathcal{R}}(\alpha)$ is not identified with a matrix nuclear norm. In particular, no $\mathfrak{S}_{\mathcal{R}}^2 \ll X^{o(1)}\mathcal{D}_0$ theorem is claimed.
- (iv) No cancellation of the coherent p -sum, no $X^{o(1)}$ residual grouping bound, and no removal of the TPC-48 coarse-edge collision factor is proved.
- (v) No sparse-alias unfolding or Fourier localization of the completed frozen field to the physical shift h_0 is proved. Only h_0 has literal physical provenance.
- (vi) The residual Gram route is not identified with the native affine physical face.
- (vii) No fixed-shift Chowla estimate, parity-barrier breach, Hardy–Littlewood asymptotic, bounded-gap theorem, or twin-prime conclusion follows.

8.3 Four next doors

The reduction leaves four logically separate continuations.

Door A: active-cell weighted survival.

Prove $\delta_w \geq X^{-1/400+o(1)}$, or a stronger subpower bound, for the literal residue/Mellin cells that actually enter (16). This requires lower information about the retained opposite-row mass, not another sup-norm estimate.

Door B: representation-frame efficiency.

Replace the Minkowski $L^1(d|\nu|)$ reassembly by a Bessel/frame estimate, or prove a quantitative no-cancellation lower Gram bound for the specified layers. This controls $c_{\mathcal{R}}$.

Door C: the coherent prime square.

Estimate, averaged over m, r , the internal sum

$$\sum_{m,r} \left\| \sum_p A_\omega(p, r, m) e(-pr\alpha) v_{\omega,p,r,m} \right\|^2 \quad (88)$$

relative to the literal prime-atomic square. This is arithmetic and remains independent of the static mask result.

Door D: localization and grouping.

Control the residual map $s = d(m)r$, the ultra-low shift interval, sparse aliases, and inversion at h_0 . This door is downstream of the first three.

Door A is now a particularly sharp next experiment: failure can be certified by an actual weighted cell of large nuclear rank, while success spends a precisely quantified part of the 1/400 budget. Door B admits an equally meaningful negative outcome if the specified layer Gram matrix has an unavoidable polynomial condition number. No failure at either door invalidates the exact static-mask theorem or the other continuation routes.

9 Deterministic certificate and reproducibility boundary

The repository contains the exact-arithmetic script

`experiments/tpc49_certificate.py.`

It produces `tpc49_certificate.json` and checks finite models of the identities used in the paper. No floating-point arithmetic or random sampling is used.

The certificate performs the following tasks.

- (i) It verifies the normalized Rademacher rank-one representation of I_n for $1 \leq n \leq 8$, its exact bounded projective mass, and the permutation nuclear/Frobenius ledger.
- (ii) It constructs complete dyadic row universes for several finite parameter sets, evaluates the literal three-factor static mask, exhausts every forbidden row type, checks the degree union bound, and finds a perfect matching by a deterministic augmenting-path algorithm.
- (iii) It verifies that the matching packet lies entirely in admitted actual-mask positions and has exact distortion equal to the row count.
- (iv) It checks the product-delta permutation on several prime fields and the exact Gram identity $A_r A_r^* = qI - J$ in a finite cyclotomic surrogate over prime fields with compatible roots of unity.
- (v) It verifies the dense Hadamard recursion, row sums, Gram matrix, number of admitted entries, and the resulting exact singular-value ledger through its two invariant subspaces.
- (vi) It checks all rational endpoint exponent differences, including $133/400 - 1/400 = 33/100$ and $267/400 - 1/400 = 133/200$.

The computation is a regression certificate, not a proof of the asymptotic row count, the nuclear theorem, or any arithmetic estimate. Those proofs are given in the text. In particular, the certificate does not search for literal prime/cofactor matching atoms and does not test a twin-prime assertion.

The committed record contains 298,427 exact checks. Its two release digests are

semantic	20c576cf7e27e0cec91cce677395a02110b857f1db53dd161f2cc12068bac3a4
source	4176a42c73c01d0b68532e1dfde30974f61f162309dcd5e001a087c0fcd9e580

The generated JSON records the check count, all exact finite parameters, a canonical semantic digest of the result payload, and the SHA-256 hash of the normalized source. Re-running the script must reproduce both hashes before a release PDF is accepted.

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